THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A polyhedron has 12 vertices and 18 edges. The number of faces is: 1

(a) 28 **(b)** 8 **(c)** 32 **(d)** 6

2. The median of 7, 18, 50, 6, 14, 44, 22, 5 is: 1

(a) 18 **(b)** 20.75 **(c)** 14 **(d)** 16

3. Think of a number n. Add 3, multiply by 5, then subtract 15. The result is: 1

(a) $5n$ **(b)** $5n + 3$ **(c)** $3n$ **(d)** $n + 3$

4. The area of a rectangle is 360 cm² and its length is 24 cm. Its breadth (in cm) is: 1

(a) 17 **(b)** 15 **(c)** 180 **(d)** 336

5. A Koch curve starts with one segment. The number of segments at stage 1 is: 1

(a) 4 **(b)** 3 **(c)** 6 **(d)** 5

6. The mean of 5 numbers is 16. If 4 of them are 12, 18, 20, 10, the remaining number is: 1

(a) 20 **(b)** 25 **(c)** 15 **(d)** 16

7. The sum of 3 consecutive integers starting with n is: 1

(a) $3n + 4$ **(b)** $3n + 3$ **(c)** $3n$ **(d)** $n + 3$

8. The area of a triangle is 72 cm² and its base is 16 cm. Its height (in cm) is: 1

(a) 9 **(b)** 12 **(c)** 11 **(d)** 18

9. In the Sierpinski triangle (area of the original triangle taken as 1), the fraction of area still shaded after stage 2 is: 1

(a) $\frac{1}{16}$ **(b)** $\frac{6}{8}$ **(c)** $\frac{8}{16}$ **(d)** $\frac{9}{16}$

10. The mean of 5 observations 39, 11, 19, 36, 15 is 24. If a new observation 12 is added, the new mean is: 1

(a) 36 **(b)** 27 **(c)** 24 **(d)** 22

11. In a number pyramid each block is the sum of the two below it. The bottom row is 1, 2, 3, 4. The top block is: 1

(a) 20 **(b)** 15 **(c)** 10 **(d)** 22

12. In a quadrilateral, a diagonal of 20 cm has perpendiculars of 6 cm and 4 cm drawn to it from the opposite vertices. Its area is: 1

(a) 30 cm² **(b)** 200 cm² **(c)** 100 cm² **(d)** 60 cm²

13. A cube of side 3 cm is painted on all faces and cut into 1 cm cubes. The number of small cubes with exactly one face painted is: 1

(a) 12 **(b)** 1 **(c)** 8 **(d)** 6

14. A plant's height (cm) at the end of weeks 1 to 6 is 4, 8, 15, 19, 22, 28. Between which two consecutive weeks was the growth the greatest? 1

(a) Between week 4 and week 5 **(b)** Between week 2 and week 3

(c) Between week 3 and week 4 **(d)** Between week 1 and week 2

15. In a calendar the nine dates in a 3×3 block have the middle date 10. Their sum is: 1

(a) 99 **(b)** 30 **(c)** 81 **(d)** 90

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. A prism has 21 edges. Find the number of sides of its base and the number of faces. 2

17. Find the median of 21, 12, 34, 44, 45, 38. 2

18. Show that the sum of 16 and its reverse is divisible by 11. 2

19. The area of a rectangular field is 1200 m² and its length is 48 m. Find its breadth and perimeter. 2

20. Verify Euler's formula for a nonagonal pyramid. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A cube of side 3 cm is painted on all faces and cut into 1 cm cubes. Find the number of small cubes with 3

(i) exactly two faces painted

(ii) exactly one face painted.

22. The runs scored by a batter in 6 innings are 78, 80, 89, 55, 31, 33. Find the mean, the median and the range. 3

23. In a number pyramid with bottom row 2, 3, 1, 4, each block is the sum of the two below it. Find all the blocks. 3

24. A rectangular lawn is 32 m by 18 m. Find its area, and the cost of levelling it at ₹ 12 per m². 3

OR

A Koch curve starts with a straight line of length 27 cm. How many segments are there at stage 3, how long is each, and what is the total length? 3

25. The marks 4, 5, 6, 7 were obtained by 3, 4, 2, 1 students respectively. Find the mean mark. 3

OR

Show that the difference of 83 and its reverse 38 is a multiple of 9. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Find the number of faces, vertices and edges of 5

(i) a triangular prism and

(ii) a triangular pyramid, and verify Euler's formula for both.

OR

The monthly rainfall (mm) from June to November is 181, 83, 44, 173, 148, 151. 5

(i) Find the mean rainfall.

(ii) Which month had the greatest rise over the previous month?

(iii) Find the range.

27. Show that the sum of 3 consecutive integers starting with n is $3n + 3$ and verify it for $n = 14$. 5

OR

A trapezium has parallel sides 20 cm and 12 cm and the distance between them is 9 cm. Find its area. A parallelogram of the same height and base 12 cm (the shorter parallel side) is cut from it, leaving a triangle: find the area left over. 5

END OF QUESTION PAPER — Total: 50 Marks

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